KYLE PELLERIN

kylepellerin@outlook.com • (603) 491-1496 • LinkedIn • Project Portfolio

EDUCATION

Bowdoin College, Brunswick, Maine **May 2026**
Bachelor of Arts in Computer Science and Environmental Studies **GPA: 3.84**
Relevant Coursework:
Computer Science: Data Structures, Algorithms, Artificial Intelligence, Systems, Distributed Systems, Deep Learning
Environmental Studies (GIS): Applications in GIS and Remote Sensing, Nature of Data, Building Resilient Communities
Awards:Sarah and James Bowdoin Scholar, Maine GIS User Group Angela Stokes Award, Bowdoin College Fritz C. A. Kollen Grant, Bowdoin Club Alpine Ski Team Captain, Sokoloff Prize, CITI Ethical Research Certifications

SKILLS AND INTERESTS

Technology: Python, Java, Swift, SwiftUI, C, ArcGIS (Pro and Online), Git, JavaScript, CSS, React.js, R, Jira, Excel
Interests: Software Development, GIS, Climate Change, AI, Community Resilience, Skiing, Agriculture, Forestry

WORK EXPERIENCE

ESRI, Redlands, California **June 2026 - September 2026**
Software Engineering Intern
- Developed an ArcGIS Pro Python toolbox and server-based Enterprise widget for rapid semantic search of Oriented Imagery, adding prompt driven object segmentation (Grounded-SAM) and automated, deduplicated feature class output.
- Selected as an Intern Hackathon Finalist and awarded People's Choice.

onX Maps, Bozeman, Montana **June 2025 - August 2025**
Software Engineering Intern
- Engineered a Swift-based authentication flow to securely authorize data requests, replaced legacy token system.
- Designed and implemented a dynamic terrain filtering tool featuring a custom SwiftUI interface, enabling users to generate and apply real-time map styling layers based on elevation, slope, and aspect criteria.
- Presented work at engineering demo days. Finished in the top 3% of the onX Summer App Challenge.

Town of Topsham Maine, Topsham, Maine **February 2025 - May 2025**
GIS Consultant
- Executed comprehensive spatial analysis using ArcGIS to update the town's key environmental health and safety and water quality indices. Delivered modernized and accurate maps to inform future zoning practices.
- Reestablished their data inventory. Presented findings and a comprehensive report at a public town hall meeting.

COBALT Team Zoestra, Portland, Maine **January 2025 - May 2025**
Software Engineering Intern: GIS Web App Design
- Designed an R Shiny-based web app to model the shifting distributions of eelgrass in Casco Bay, Maine, over the past 30 years to assist with community-based conservation and research efforts.
- Collaborated with stakeholders to create and implement a data upload system for the app, which employs user-collected field data to accurately monitor real-time dynamics in the eelgrass beds.

Bowdoin College, Brunswick, Maine
GIS Teaching Assistant **August 2025 - May 2026**
- Held office hours weekly, and designed ArcGIS projects for the *Building Resilient Communities* GIS Course.

Research Assistant **October 2023 - May 2026**
- Established a quantitative data analysis metric for evaluating the effectiveness of the Maine Governor's Office of Policy and Innovation for the Future's Community Resilience Program.
- Researched the relationship between municipalities' ability to apply for grants, expand their digital capacity, and increase community resilience. Presented findings at the *American Association of Geographers* conference.

PROJECTS

Assessing Geospatial Modeling in Determining Optimal Afforestation Locations for Carbon Sequestration
- Produced a modeling tool using ArcGIS Pro to facilitate the government-funded afforestation in Isafjord, Iceland, with a specific focus on optimizing the potential for future carbon sequestration.
- Authored the above paper that located two ideal planting sites, which are now used as afforestation zones.

S&P Finance
- Collaborated with a peer to design and author a web application using JavaScript, CSS, and the React.js framework to provide financial information to students. Launched the website using AWS.